LUNGS

THE TWO SPONGE-LIKE LUNGS FILL MOST OF THE CHEST CAVITY AND ARE PROTECTED BY THE RIBS. THEIR ESSENTIAL FUNCTION IS GAS EXCHANGE – TAKING IN VITAL OXYGEN FROM THE AIR AND EXPELLING WASTE CARBON DIOXIDE.

LUNG STRUCTURE

Air enters the lungs from the trachea, which branches at its base into two main airways, the primary bronchi. Each primary bronchus enters its lung at a site called the hilum, which is also where the main blood vessels pass in and out of the lung. The primary bronchus divides into secondary bronchi, and these subdivide into tertiary bronchi, all the time decreasing in diameter. Many subsequent divisions form the narrowest airways: the terminal and then respiratory bronchioles, which distribute air to the alveoli.

ALVEOLI

The lungs' microscopic air sacs, known as alveoli, are elastic, thin-walled structures arranged in clumps at the ends of respiratory bronchioles. Around the alveoli are networks of capillaries. Oxygen passes from the air in the alveoli into the blood by diffusion through the alveolar and capillary walls (see p.166). Carbon dioxide diffuses from the blood into the alveoli. There are more than 300 million alveoli in both lungs, providing a huge surface area for gas exchange.